



PURELIVINGZONE · FREE GUIDE

The Water Safety Card

Exact amounts and times for every purification method — small enough for your wallet

Three methods. Known for a hundred years. Everything you need to make any water safe to drink — the exact amounts, exact times, and what not to do.

METHOD 1 — BOILING (MOST RELIABLE)

- **Bring water to a full rolling boil** — not simmering, not steaming. Rolling means large rapid bubbles that cannot be stirred down.
 - **Boil for 1 minute** — this is sufficient at most elevations to kill bacteria, viruses, and protozoa (including Giardia and Cryptosporidium).
 - **At altitude above 2,000 m (6,500 ft)** — boil for 3 minutes. Water boils at a lower temperature at altitude; the extra time compensates.
 - **Cool with a cover on** — place a lid or cloth over the container while it cools. An uncovered pot can be re-contaminated by insects, dust, or hands.
 - **Boiled water tastes flat** — pour it between two containers several times to re-aerate it. Taste improves significantly.
 - **Does not remove** — chemicals, heavy metals, or salt. Boiling is for biological contamination only.
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METHOD 2 – SODIS (SOLAR DISINFECTION – NO FIRE REQUIRED)

- **Clear PET plastic bottle only** — the label should have a recycling symbol with 'PET' or '1' inside. Glass also works. Colored, scratched, or opaque bottles reduce effectiveness significantly.
 - **Fill the bottle. Shake it for 20 seconds** — this adds dissolved oxygen, which accelerates the disinfection process.
 - Lay the bottle flat in full direct sunlight. A metal or foil-covered surface below the bottle increases reflected UV exposure.
 - Leave for 6 hours in full bright sunshine. Leave for 12 hours if the sky is more than 50% cloudy.
 - **Do NOT use if** — the water is turbid (brown, murky). Suspended particles shield pathogens from UV light. Pre-filter turbid water first (see below).
 - **Works by** — UV-A radiation damages the DNA of microorganisms, preventing reproduction. Effective against bacteria, viruses, and protozoa.
 - **Proven field use** — documented as effective by WHO in humanitarian settings across Africa and Asia.
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METHOD 3 – HOUSEHOLD BLEACH (FASTEST, MOST PORTABLE)

- **Use unscented sodium hypochlorite bleach only** — 5–8% concentration. Check the label. Scented, color-safe, or 'ultra' thick bleaches contain other chemicals not safe for drinking water.
 - **Dose** — 8 drops per gallon (3.8 L) of clear water. For a liter: 2 drops. For a 2-liter bottle: 4 drops.
 - **If water is cloudy or cold** — double the dose (16 drops per gallon / 4 drops per liter).
 - Stir. Wait 30 minutes.
 - **The smell test** — after 30 minutes, the water should have a faint bleach smell. This confirms the chlorine is still active and effective.
 - **No smell after 30 minutes** — add the same dose again. Stir. Wait another 15 minutes before drinking.
 - **Strong bleach smell after 30 minutes** — the dose was too high, but the water is safe. Allow it to stand uncovered for a few minutes; the excess chlorine will off-gas.
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PRE-FILTERING TURBID (CLOUDY) WATER

- Any method is significantly less effective in turbid (brown, silty, or murky) water. Filter before treating.
 - **Cloth filter** — pour water through multiple layers of clean cloth (cotton T-shirt folded 4 times). Removes large particles.
 - **Settling** — let the water sit undisturbed for 1–2 hours. Large particles sink. Pour off the clearer top layer carefully.
 - **Sand filter** — a clean container with layers of gravel on the bottom, then sand, then fine sand or cloth on top. Pour water through. Effective for field use.
 - **After filtering** — the water is still biologically unsafe. Apply one of the three methods above before drinking.
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WATER STORAGE

- **Treated water** — store in a sealed, clean container in a cool, dark place. Consume within 24–48 hours if the container is opened, or up to 6 months if sealed and untouched.
 - **Containers** — food-grade plastic, stainless steel, or glass. Never galvanized metal (zinc leaches into water) or non-food-grade plastic.
 - **Minimum reserve** — keep at least 3 liters per person per day for a minimum of 3 days in your home. FEMA recommends 1 gallon (3.8 L) per person per day.
 - **Rotation** — use and replenish stored water every 6–12 months. Taste degrades in older stored water, not safety.
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WHAT DOES NOT WORK

- **Boiling salt water** — removes biological contamination but leaves the salt. Drinking concentrated salt water accelerates dehydration and causes organ damage. Only distillation removes salt.
 - **UV pens (SteriPen)** — effective only in clear water. Will not work in turbid water. Requires batteries. Not a substitute when these conditions are not met.
 - **Iodine tablets alone** — effective against bacteria and viruses but NOT against *Cryptosporidium*. Use only when no other method is available.
 - **Filtering through sand alone** — removes particles but not dissolved viruses and bacteria. Always combine with a chemical or UV treatment.
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KEEP THIS IN MIND

Print this page, fold it to credit-card size, laminate it or wrap it in tape, and put it in your wallet. The man who has already read this page is calm when others panic.

*This is the last guide in this week's series. Share it with one person who would want it.
The prepared man does not guard knowledge — he passes it on.*

SOURCES World Health Organization — Household Water Treatment and Safe Storage · U.S. Centers for Disease Control and Prevention (CDC) — Making Water Safe in an Emergency · FEMA — Water in an Emergency (ready.gov) · Acra A. et al. — 'Sunlight as disinfectant,' The Lancet, 1984 (foundational SODIS research) · WHO/UNICEF — SODIS Field Manual for Household Water Treatment

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